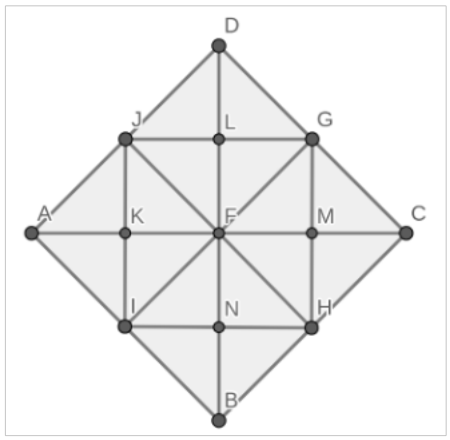
Grade 8
Unit 9
Lesson 10 Practice

Name_Answer Key________
Date____________________
Period__________________

The Louvre in Paris includes a skylight in the Carrousel du Louvre shopping mall in front of the Louvre Museum. This skylight is the Inverted Pyramid, the square base of which is a walkway to direct visitors towards the museum entrance. The museum wants to create a second larger Inverted Pyramid, the base of which is represented below.

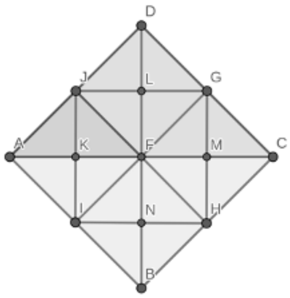
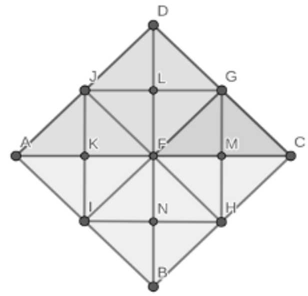
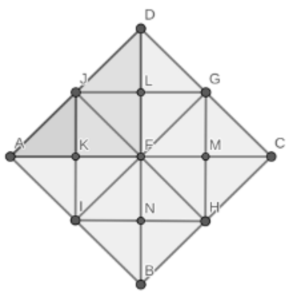
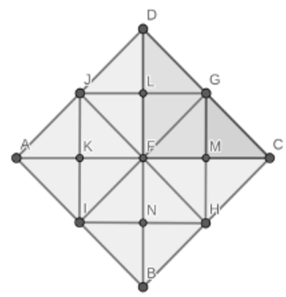


The side lengths of the new square base are 66 feet (ft). Steel beams are placed from the center of the square base and attached perpendicularly to each side of the base. These beams help reinforce the strength of the pyramid as visitors walk above it.

Use the Pythagorean Theorem and other reasoning to find the lengths of the line segments. Round to the nearest tenth of a foot.

Line Segment	Work/Reasoning	Length (ft)
1. $\overline{AD}$	Given	66
2. $\overline{DC}$	Given	66
3. $\overline{AC}$	Base of Right $\triangle ADC$ $66^2 + 66^2 = \overline{AC}^2$ $AC^2 = 8{,}712 \rightarrow\ AC = 93.3$	93.3
4. $\overline{DF}$	$AC = DB$ $DF = \frac{1}{2}DB \rightarrow DF = \frac{1}{2}(93.3)$ $DF = 46.7$	46.7

The images and statements below give additional information about the Inverted Pyramid design.

 $\triangle ADC$ is a dilation of $\triangle AJF$ using a scale factor of 2 centered at point A.	 $\triangle ADC$ is a dilation of $\triangle FGC$ using a scale factor of 2 centered at point C.
 $\triangle AJK$ is a dilation of $\triangle ADF$ using a scale factor of $\frac{1}{2}$ centered at point A.	 $\triangle CGM$ is a dilation of $\triangle CDF$ using a scale factor of $\frac{1}{2}$ centered at point C.

Use the Pythagorean Theorem, properties of similar triangles and other reasoning to find the lengths of the line segments. Round to the nearest tenth of a foot.

Line Segment	Work/Reasoning	Length (ft)
5. $\overline{AJ}$	Image $\triangle ADC$, preimage $\triangle AJF$ $\frac{AD}{AJ} = 2 \rightarrow \frac{66}{AJ} = 2$ $66 \div 2 = 33 \rightarrow \overline{AJ} = 33$	33
6. $\overline{JF}$	$\overline{AD} = \overline{DC}$ so $\overline{AJ} = \overline{JF}$ $\overline{JF} = 33$	33
7. $\overline{AF}$	$\overline{AC} = 93.3$ $\frac{AC}{AF} = 2 \rightarrow \frac{93.3}{AF} = 2$ $93.3 \div 2 = 46.7 \rightarrow \overline{AF} = 46.7$	46.7
8. $\overline{GF}$	$\frac{AD}{GF} = 2 \rightarrow \frac{66}{GF} = 2$ $66 \div 2 = 33 \rightarrow \overline{GF} = 33$	33

9. $\overline{AK}$	$\overline{AF} = 46.7 \rightarrow \frac{AK}{46.7} = \frac{1}{2}$ $46.7 \cdot \frac{1}{2} = 23.4 \rightarrow \overline{AK} = 23.4$	23.4
10. $\overline{GM}$	$DF = 46.7 \rightarrow \frac{GM}{DF} = \frac{1}{2} \rightarrow \frac{GM}{46.7} = \frac{1}{2}$ $46.7 \cdot \frac{1}{2} = 23.4 \rightarrow \overline{GM} = 23.4$	23.4